
EXERCISES

Homework 2

AI503: Calculus

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1 Problem 1

A company's profit function is given by

$$P(x) = -2x^2 + 12x - 5$$

where x is the number of units produced. Find the production level that maximizes profit.

1.1 Solution

Find $f'(x)$

$$f'(x) = -4x + 12$$

Then

$$0 = -4x + 12 \Rightarrow x = \frac{12}{4} = 3$$

Since $f''(x) = -4 < 0$ graph is a concave down everywhere

Because of this $x = 3$ must be the maximum

2 Problem 2

Let $\vec{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$. Compute:

1. $\vec{u} + \vec{v}$
2. $2\vec{u} - 3\vec{v}$
3. $\vec{u} \cdot \vec{v}$

2.1 Solution

1. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 5 \end{pmatrix}$
2. $2\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} - \begin{pmatrix} 12 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -10 \\ 7 \\ 0 \end{pmatrix}$
3. $\vec{u} \cdot \vec{v} = 1 \cdot 4 + 2 \cdot (-1) + 3 \cdot 2 = 8$

3 Problem 3

Find the length (norm) of the vector $\vec{w} = \begin{pmatrix} 2 \\ -1 \\ 2 \\ 2 \end{pmatrix}$. And normalize it to a unit vector.

3.1 Solution

The length of vector is define as

$$\|\vec{w}\| = \sqrt{2^2 + (-1)^2 + 2^2 + 2^2} = \sqrt{13}$$

To normalize $\vec{w}$, set the length to 1 by: $\vec{w}_u = \frac{1}{\sqrt{13}} \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{13}} \\ -\frac{1}{\sqrt{13}} \\ \frac{2}{\sqrt{13}} \\ \frac{2}{\sqrt{13}} \end{pmatrix}$

$$\text{Now } \|\vec{w}\| = \sqrt{\left(\frac{2}{\sqrt{13}}\right)^2 + \left(-\frac{1}{\sqrt{13}}\right)^2 + \left(\frac{2}{\sqrt{13}}\right)^2 + \left(\frac{2}{\sqrt{13}}\right)^2} = 1$$

4 Problem 4

Let $a = (1, 0, -1)$ and $b = (2, 3, 1)$. Compute the angle between a and b .

4.1 Solution

Dot product is defined as

$$a \cdot b = \|a\| \|b\| \cdot \underbrace{\cos(\theta)}_{\angle(a,b)}$$

So.

$$\theta = \cos^{-1} \left(\frac{a \cdot b}{\|a\| \|b\|} \right)$$

Now to find θ between a and b

$$\theta = \cos^{-1} \left(\frac{2 - 1}{\sqrt{2} \cdot \sqrt{14}} \right)$$

5 Problem 5

Find a linear function whose graph is the plane intersects the xy -plane along the line $y = 2x + 2$ and contains the point $(1, 2, 2)$

5.1 Solution

The plane equation in 3d:

$$ax + by + cz = d$$

Where $y = 2x + 2$ and z is fixed to $z = 0$ at

So then

$$ax + b(2x + 2) + c(0) = ax + 2bx + b2 = d$$

This can be rewritten to

$$(a + 2b)x + b2 = d$$

Since d is fixed: $a + 2b = 0$, $a = -2b$ and $d = 2b$

Now to choose some arbitrary b -value $b = 1$ (for simplicity)

Then

$$-2x + y + cz = 2$$

This plane is not fixed in the point yet, lets do that and find c to fix the plane to the point

$$(1, 2, 2) \Rightarrow -2(1) + 2 + c2 = 2 \Rightarrow c2 = 2 \Rightarrow c = 1$$

Finally

$$\mathbf{-2x + y + z = 2}$$

6 Problem 6

Determine if z is a function of x and y :

1. $6x - 4y + 2z = 10$
2. $x^2 + y^2 + z^2 = 100$
3. $3x^2 - 5y^2 + 5x = 10 + x + y$

6.1 Solution

1.
$$\begin{aligned} 6x - 4y + 2z &= 10 \\ 2z &= -6x + 4y + 10 \\ z &= -3x + 2y + 5 \quad \checkmark \end{aligned}$$

2.
$$\begin{aligned} x^2 + y^2 + z^2 &= 100 \\ z^2 &= 100 - x^2 - y^2 \\ z &= \sqrt{100 - x^2 - y^2} \quad \checkmark \end{aligned}$$

3.
$$\begin{aligned} 3x^2 - 5y^2 + 5x &= 10 + x + y \\ 5z &= -3x^2 - 5y^2 + 10 + x + y \\ z &= \frac{-3x^2 - 5y^2 + 10 + x + y}{5} \quad \checkmark \end{aligned}$$

7 Problem 7

Show that the following limit does not exist

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x + y^2}{2x + y}$$

7.1 Solution

Try different paths $x = 0$ and $y = 0$

First $x = 0$

$$\lim_{x \rightarrow 0} \frac{x}{2x} = \frac{1}{2}$$

Then $y = 0$

$$\lim_{y \rightarrow 0} \frac{y^2}{y} = 0$$

Since $\frac{1}{2} \neq 0$ The limit does not exist

8 Problem 8

Find the following limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{\sin(y)+2}$$

8.1 Solution

Try different paths $x = 0$ and $y = 0$

First $x = 0$

$$\lim_{y \rightarrow 0} \frac{y}{2} = 0$$

Then $y = 0$

$$\lim_{x \rightarrow 0} \frac{x}{2} = 0$$